

Adam: A Method for Stochastic Optimization

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Abstract

We introduce Adam, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyper-parameters have intuitive interpretations and typically require little tuning.

1. Introduction

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems in these fields can be cast as the optimization of some scalar parameterized objective function requiring maximization or minimization with respect to its parameters. We introduce Adam, a method for efficient stochastic optimization that only requires first-order gradients with little memory requirement. The method computes individual adaptive learning rates for different parameters from estimates of first and second moments of the gradients.

2. Algorithm

The name Adam is derived from adaptive moment estimation. Adam is designed to combine the advantages of two recently popular methods: AdaGrad, which works well with sparse gradients, and RMSProp, which works well in online and non-stationary settings. Adam computes exponential moving averages of the gradient (the first moment) and the squared gradient (the second moment), with hyper-parameters β_1 , β_2 controlling the decay rates of these moving averages.

$$\text{Eq: } m_t = \beta_1 * m_{t-1} + (1 - \beta_1) * g_t$$

$$\text{Eq: } v_t = \beta_2 * v_{t-1} + (1 - \beta_2) * g_t^2$$

The moving averages m_t and v_t are estimates of the first moment (the mean) and the second raw moment (the uncentered variance) of the gradient respectively. Because m_t and v_t are initialized as vectors of zeros, they are biased towards zero, especially during the initial timesteps. We counteract these biases by computing bias-corrected estimates before the parameter update.

Good default settings for the tested machine learning problems are $\alpha=0.001$, $\beta_1=0.9$, $\beta_2=0.999$ and $\epsilon=1e-8$.

4. Experiments

We compared Adam to other stochastic optimization methods on logistic regression, multi-layer neural networks, and convolutional neural networks. Empirical results demonstrate that Adam works well in practice and compares favorably to other stochastic optimization methods, including AdaGrad, RMSProp, and SGD with Nesterov momentum.

6. Conclusion

We have introduced a simple and computationally efficient algorithm for gradient-based optimization of stochastic objective functions. Our method is aimed towards machine learning problems with large datasets and/or high-dimensional parameter spaces. Overall, we found Adam to be robust and well-suited to a wide range of non-convex optimization problems in the field of machine learning.